

NBA Game Win Probability Predictor

A distributed ML pipeline for predicting NBA game outcomes using historical team performance data.

<https://sthomas10-nba-predictor.hf.space/>

PROJECT OVERVIEW

This project implements an end-to-end distributed data pipeline and machine learning system that predicts the win probability for any NBA matchup. Given a home and away team, the system returns real-time win probabilities derived from each team's rolling 10-game performance averages and contextual factors such as home-court advantage and rest days.

DATA PIPELINE ARCHITECTURE

Bronze Layer — Raw game logs and team metrics are ingested from the NBA Stats API using the `nba_api` Python library and stored as JSON in Amazon S3 (`s3://nbapredictions-sthomas26-ncf/bronze/`).

Silver Layer — PySpark on Databricks Community Edition reads the raw JSON, casts and cleans all fields, drops null records, and writes typed Parquet files back to S3 (`silver/games/`).

Gold Layer — A second PySpark job applies feature engineering: rolling 10-game averages for points, rebounds, assists, turnovers, and plus-minus; home/away flags derived from matchup strings; and rest days computed via window functions. The output is a model-ready Parquet table (`gold/games/`).

MODEL TRAINING

An XGBoost binary classifier (`n_estimators=100`, `max_depth=4`, `learning_rate=0.1`) was trained on the gold layer features to predict home team win (`WIN=1`) or loss (`WIN=0`). The dataset covers the full 2024-25 NBA regular season with an 80/20 train-test split. All experiments were tracked using MLflow on Databricks, with parameters, metrics, and the serialized model artifact logged to the MLflow experiment registry.

FEATURE SET

Feature	Description
HOME	1 if home team, 0 if away
REST_DAYS	Days since last game
AVG_PTS_L10	Avg points, last 10 games

AVG_REB_L10	Avg rebounds, last 10 games
AVG_AST_L10	Avg assists, last 10 games
AVG_TOV_L10	Avg turnovers, last 10 games
AVG_PLUS_MINUS_L10	Avg plus/minus, last 10 games

MODEL RESULTS

Metric	Value
Accuracy	58.5%
AUC-ROC	0.633
Algorithm	XGBoost
Train/Test Split	80% / 20%
Season	2024-25 NBA

SERVING & DEPLOYMENT

The trained model is serialized and stored in S3 (`model/nba_model.pkl`). A FastAPI application loads the model on startup, reads the latest team rolling averages from the gold layer, and exposes a `POST /predict` endpoint. The API is containerized via Docker and deployed on Hugging Face Spaces, providing a persistent public URL. A React-inspired HTML/CSS frontend renders team selectors and animated win-probability bars.

DISCUSSION

An accuracy of 58.5% and AUC of 0.633 exceed the 50% random baseline, consistent with published NBA prediction benchmarks (55-65% range). Performance is bounded by the inherent variability of basketball — injuries, lineup changes, and in-game momentum are not captured by rolling averages alone. Future improvements could incorporate player-level data, opponent-adjusted metrics, and travel fatigue.